

Operations Research II: Algorithms

The Simplex Method

Ling-Chieh Kung

Department of Information Management
National Taiwan University

Introduction

- ▶ Let's study how to **solve** an LP.
- ▶ The algorithm we will introduce is **the simplex method**.
 - ▶ Developed by **George Dantzig** in 1947.
 - ▶ Opened the whole field of Operations Research.
 - ▶ Implemented in most commercial LP solvers.
 - ▶ **Very efficient** for almost all practical LPs.
 - ▶ With **very simple ideas**.
- ▶ The method is general in an indirect manner.
 - ▶ There are many different forms of LPs.
 - ▶ We will first show that each LP is equivalent to a **standard form** LP.
 - ▶ Then we will show how to solve standard form LPs.
- ▶ This lecture will be full of **algebra** and **theorems**. Get ready!

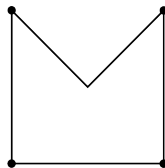
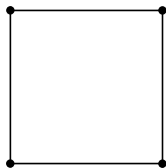
Extreme points

- We need to first define **extreme points** for a set:

Definition 1 (Extreme points)

For a set $S \subseteq \mathbb{R}^n$, a point x is an extreme point if there does not exist a three-tuple (x^1, x^2, λ) such that $x^1 \in S \setminus \{x\}$, $x^2 \in S \setminus \{x\}$, $\lambda \in (0, 1)$, and

$$x = \lambda x^1 + (1 - \lambda)x^2.$$



Optimality of extreme points

- ▶ For any LP, we have the following fact.

Proposition 1

For any LP, if there is an optimal solution, there is an extreme point optimal solution.

- ▶ It is not saying that “if a solution is optimal, it is an extreme point!”
- ▶ This property will be very useful when we develop a method for solving general LPs!

Road map

- ▶ **The standard form.**
- ▶ Basic solutions.
- ▶ The simplex method.
- ▶ The tableau representation.
- ▶ Unbounded LPs.
- ▶ Infeasible LPs.

Standard form LPs

- ▶ First, let's define the **standard form**.

Definition 2 (Standard form LP)

An LP is in the standard form if

- ▶ *all the RHS values are nonnegative,*
- ▶ *all the variables are nonnegative, and*
- ▶ *all the constraints are equalities.*

- ▶ RHS = right hand sides. For any constraint

$$g(x) \leq b, \quad g(x) \geq b, \quad \text{or} \quad g(x) = b,$$

b is the RHS.

- ▶ There is no restriction on the objective function.

Finding the standard form

- ▶ How to find the standard form for an LP?
- ▶ Requirement 1: **Nonnegative RHS**.
 - ▶ If it is negative, **switch** the LHS and the RHS.
 - ▶ E.g.,

$$2x_1 + 3x_2 \leq -4$$

is equivalent to

$$-2x_1 - 3x_2 \geq 4.$$

Finding the standard form

► Requirement 2: **Nonnegative variables.**

- If x_i is **nonpositive**, replace it by $-x_i$. E.g.,

$$2x_1 + 3x_2 \leq 4, x_1 \leq 0 \quad \Leftrightarrow \quad -2x_1 + 3x_2 \leq 4, x_1 \geq 0.$$

- If x_i is **free**, replace it by $x'_i - x''_i$, where $x'_i, x''_i \geq 0$. E.g.,

$$2x_1 + 3x_2 \leq 4, x_1 \text{ free.} \quad \Leftrightarrow \quad 2x'_1 - 2x''_1 + 3x_2 \leq 4, x'_1 \geq 0, x''_1 \geq 0.$$

$x_i = x'_i - x''_i$	$x'_i \geq 0$	$x''_i \geq 0$
5	5	0
0	0	0
-8	0	8

Finding the standard form

► Requirement 3: **Equality constraints.**

- For a “ $\leq$ ” constraint, **add a slack** variable. E.g.,

$$2x_1 + 3x_2 \leq 4 \quad \Leftrightarrow \quad 2x_1 + 3x_2 + x_3 = 4, \quad x_3 \geq 0.$$

- For a “ $\geq$ ” constraint, **minus a surplus/excess** variable. E.g.,

$$2x_1 + 3x_2 \geq 4 \quad \Leftrightarrow \quad 2x_1 + 3x_2 - x_3 = 4, \quad x_3 \geq 0.$$

- For ease of exposition, they will both be called slack variables.
► A slack variable measures the **gap** between the LHS and RHS.

An example

$$\begin{array}{llllll}
 \min & 3x_1 & + & 2x_2 & + & 4x_3 \\
 \text{s.t.} & x_1 & + & 2x_2 & - & x_3 & \geq & 6 \\
 & x_1 & - & x_2 & & & \geq & -8 \\
 & 2x_1 & + & x_2 & + & x_3 & = & 9 \\
 & x_1 \geq 0, & x_2 \leq 0, & x_3 \text{ urs.}
 \end{array}$$

$$\begin{array}{llllllllll}
 \min & 3x_1 & - & 2x_2 & + & 4x_3 & - & 4x_4 & & & \\
 \rightarrow \text{s.t.} & x_1 & - & 2x_2 & - & x_3 & + & x_4 & - & x_5 & = & 6 \\
 & -x_1 & - & x_2 & & & & & & + & x_6 & = & 8 \\
 & 2x_1 & - & x_2 & + & x_3 & - & x_4 & & & = & 9 \\
 & x_i \geq 0 & \forall i = 1, \dots, 6.
 \end{array}$$

Standard form LPs in matrices

- ▶ Given **any** LP, we may find its standard form.
- ▶ With matrices, a standard form LP is expressed as

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned}$$

- ▶ E.g., for

$$\begin{aligned} \min \quad & 2x_1 - x_2 \\ \text{s.t.} \quad & x_1 + 5x_2 + x_3 = 5 \\ & 3x_1 - 6x_2 + x_4 = 4 \\ & x_i \geq 0 \quad \forall i = 1, \dots, 4, \end{aligned}$$

$$c = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 0 \end{bmatrix}, b = \begin{bmatrix} 5 \\ 4 \end{bmatrix}, \text{ and } A = \begin{bmatrix} 1 & 5 & 1 & 0 \\ 3 & -6 & 0 & 1 \end{bmatrix}.$$

- ▶ We will denote the number of constraints and variables as m and n .
 - ▶ $A \in \mathbb{R}^{m \times n}$ is called the **coefficient matrix**.
 - ▶ $b \in \mathbb{R}^m$ is called the **RHS vector**.
 - ▶ $c \in \mathbb{R}^n$ is called the **objective vector**.
- ▶ The objective function can be either max or min.

Solving standard form LPs

- ▶ So now we only need to find a way to solve standard form LPs.
- ▶ How?
- ▶ A standard form LP is still an LP.
- ▶ If it has an optimal solution, it has an **extreme point** optimal solution! Therefore, we only need to search among extreme points.

Road map

- ▶ The standard form.
- ▶ **Basic solutions.**
- ▶ The simplex method.
- ▶ The tableau representation.
- ▶ Unbounded LPs.
- ▶ Infeasible LPs.

Basic solutions

- ▶ Consider a standard form LP with m constraints and n variables

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax = b \\ & x \geq 0.\end{array}$$

- ▶ We may assume that A has m pivots, i.e., all rows of A are independent.¹
- ▶ This then implies that $m \leq n$. As the problem with $m = n$ is trivial, we will assume that $m < n$.

¹This assumption is without loss of generality. Why?

Basic solutions

- ▶ For the system $Ax = b$, now there are more columns than rows. Let's select some columns to form a **basic solution**:

Definition 3 (Basic solution)

A basic solution to a standard form LP is a solution that (1) has $n - m$ variables being equal to 0 and (2) satisfies $Ax = b$.

- ▶ The $n - m$ variables chosen to be zero are **nonbasic variables**.
- ▶ The remaining m variables are **basic variables**.
- ▶ The set of basic variables is called a **basis**.
- ▶ These m columns form a nonsingular/invertible $m \times m$ matrix A_B .
- ▶ We use $x_B \in \mathbb{R}^m$ and $x_N \in \mathbb{R}^{n-m}$ to denote basic and nonbasic variables, respectively, with respect to a given set of basic variables B .
 - ▶ We have $x_N = 0$ and $x_B = A_B^{-1}b$.

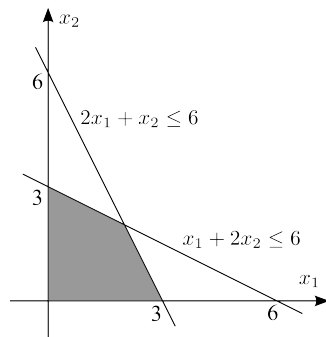
Basic solutions: an example

- Consider an original LP

$$\begin{array}{llllll} \min & 6x_1 & + & 8x_2 & & \\ \text{s.t.} & x_1 & + & 2x_2 & \leq & 6 \\ & 2x_1 & + & x_2 & \leq & 6 \\ & x_i & \geq & 0 & \forall i = 1, 2 \end{array}$$

and its standard form

$$\begin{array}{llllllll} \min & 6x_1 & + & 8x_2 & & & & \\ \text{s.t.} & x_1 & + & 2x_2 & + & x_3 & = & 6 \\ & 2x_1 & + & x_2 & & & + & x_4 = 6 \\ & x_i & \geq & 0 & \forall i = 1, \dots, 4. \end{array}$$



Basic solutions: an example

- ▶ In the standard form, $m = 2$ and $n = 4$.
 - ▶ There are $n - m = 2$ nonbasic variables.
 - ▶ There are $m = 2$ basic variables.
- ▶ Steps for obtaining a basic solution:
 - ▶ Determine a set of m basic variables to form a basis B .
 - ▶ The remaining variables form the set of nonbasic variables N .
 - ▶ Set nonbasic variables to zero: $x_N = 0$.
 - ▶ Solve the m by m system $A_B x_B = b$ for the values of basic variables.
- ▶ For this example, we will solve a two by two system for each basis.

Basic solutions: an example

- The two equalities are

$$\begin{array}{rcccccccl} x_1 & + & 2x_2 & + & x_3 & & & = & 6 \\ 2x_1 & + & x_2 & & & + & x_4 & = & 6. \end{array}$$

- Let's try $B = (x_1, x_2)$ and $N = (x_3, x_4)$:

$$\begin{array}{rcccl} x_1 & + & 2x_2 & = & 6 \\ 2x_1 & + & x_2 & = & 6. \end{array}$$

The solution is $(x_1, x_2) = (2, 2)$. Therefore, the basic solution associated with this basis B is $(x_1, x_2, x_3, x_4) = (2, 2, 0, 0)$.

- Let's try $B = (x_2, x_3)$ and $N = (x_1, x_4)$:

$$\begin{array}{rcccl} 2x_2 & + & x_3 & = & 6 \\ x_2 & & & = & 6. \end{array}$$

As $(x_2, x_3) = (6, -6)$, the basic solution is $(x_1, x_2, x_3, x_4) = (0, 6, -6, 0)$.

Basic solutions: an example

- ▶ In general, as we need to choose m out of n variables to be basic, we have **at most** $\binom{n}{m}$ different bases.²
- ▶ In this example, we have exactly $\binom{4}{2} = 6$ bases.
- ▶ By examining all the six bases one by one, we may find all those associated basic variables:

B	Basic solution			
	x_1	x_2	x_3	x_4
(x_1, x_2)	2	2	0	0
(x_1, x_3)	3	0	3	0
(x_1, x_4)	6	0	0	-6
(x_2, x_3)	0	6	-6	0
(x_2, x_4)	0	3	0	3
(x_3, x_4)	0	0	6	6

²Why “at most”? Why not “exactly”?

Basic feasible solutions

- ▶ Among all basic solutions, some are feasible.
 - ▶ By the definition of basic solutions, they satisfy $Ax = b$.
 - ▶ If one also **satisfies** $x \geq 0$, it satisfies all constraints.
- ▶ In this case, it is called **basic feasible solutions** (bfs).

Definition 4 (Basic feasible solution)

A basic feasible solution to a standard form LP is a basic solution whose basic variables are all nonnegative.

- ▶ Which are bfs?

Basis	Basic solution			
	x_1	x_2	x_3	x_4
(x_1, x_2)	2	2	0	0
(x_1, x_3)	3	0	3	0
(x_1, x_4)	6	0	0	-6
(x_2, x_3)	0	6	-6	0
(x_2, x_4)	0	3	0	3
(x_3, x_4)	0	0	6	6

Basic feasible solutions and extreme points

- ▶ Why bfs are important? They are just extreme points!

Theorem 1 (Extreme points and basic feasible solutions)

For a standard form LP, a solution is an extreme point of the feasible region if and only if it is a basic feasible solution to the LP.

- ▶ The implication is direct:

Theorem 2 (Optimality of basic feasible solutions)

For a standard form LP, if there is an optimal solution, there is an optimal basic feasible solution.

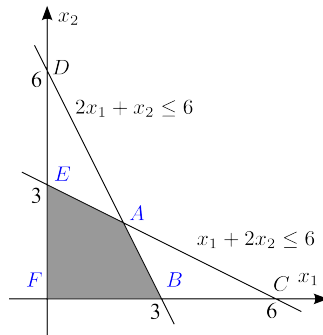
- ▶ Though we cannot prove Theorem 1 here, let's get some intuitions.³

³Please note that these “intuitions” are never rigorous.

An example

- There is a one-to-one mapping between bfs and extreme points.

Basis	Bfs?	Point	Basic solution			
			x_1	x_2	x_3	x_4
(x_1, x_2)	Yes	<i>A</i>	2	2	0	0
(x_1, x_3)	Yes	<i>B</i>	3	0	3	0
(x_1, x_4)	No	<i>C</i>	6	0	0	-6
(x_2, x_3)	No	<i>D</i>	0	6	-6	0
(x_2, x_4)	Yes	<i>E</i>	0	3	0	3
(x_3, x_4)	Yes	<i>F</i>	0	0	6	6



Solving standard form LPs

- ▶ To find an optimal solution:
 - ▶ Instead of searching among all extreme points, we search among **all bfs**.
 - ▶ Extreme points are defined **geometrically**; bfs are **algebraically**.
 - ▶ Checking whether a solution is basic feasible is easy (for a computer).
- ▶ To search among bfs, we keep moving to a better **adjacent** bfs from the current one:

Definition 5 (Adjacent bases and bfs)

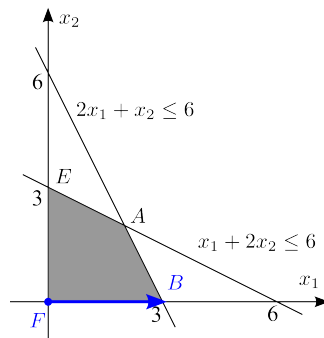
*Two bases are adjacent if exactly one of their variables is different.
Two bfs are adjacent if their associated bases are adjacent.*

- ▶ Again, let's use a graph to get the idea.

Adjacent basic feasible solutions

- ▶ A pair of adjacent bfs corresponds to a pair of “adjacent” extreme points, i.e., extreme points that are on **the same edge**.
- ▶ Switching from a bfs to its adjacent bfs is **moving along an edge**.

Basis	Point	Basic solution			
		x_1	x_2	x_3	x_4
(x_1, x_2)	A	2	2	0	0
(x_1, x_3)	B	3	0	3	0
(x_2, x_4)	E	0	3	0	3
(x_3, x_4)	F	0	0	6	6



A better way to search

- ▶ Given all these concepts, how would you search among bfs?
- ▶ At each bfs, move to an **adjacent** bfs that is **better**!
 - ▶ Around the current bfs, there should be some improving directions.
 - ▶ Otherwise, the bfs is optimal.
- ▶ Next we will introduce the simplex method, which utilize this idea in an elegant way.

Road map

- ▶ The standard form.
- ▶ Basic solutions.
- ▶ **The simplex method.**
- ▶ The tableau representation.
- ▶ Unbounded LPs.
- ▶ Infeasible LPs.

The idea

- ▶ All we need is to search among bfs.
 - ▶ Geometrically, we search among extreme points.
 - ▶ Moving to an adjacent bfs is to move along an edge.
- ▶ Questions:
 - ▶ Which edge to move along?
 - ▶ When to stop moving?
- ▶ All these must be done with algebra rather than geometry.
- ▶ Algebraically, to move to an adjacent bfs, we need to **replace** one basic variable by a nonbasic variable.
 - ▶ E.g., moving from $B_1 = (x_1, x_2, x_3)$ to $B_2 = (x_2, x_3, x_5)$.
- ▶ There are two things to do:
 - ▶ Select one **nonbasic** variable to **enter** the basis, and
 - ▶ Select one **basic** variable to **leave** the basis.

The idea

- ▶ Entering and leaving:
 - ▶ Selecting one nonbasic variable to **enter** means making it **nonzero**: Increasing its value from 0 to a positive value and become **basic**.
 - ▶ While this variable increases, we identify basic variables that decrease and stop when one hits 0. That variable **leaves** the basis and become **nonbasic**.
- ▶ We keep **changing the basis** until we find an optimal basis.
- ▶ Next let's know exactly how to run the simplex method in algebra.

The simplex method

- To introduce the algebra of the simplex method, let's consider the following LP

$$\begin{array}{ll} \max & 2x_1 + 3x_2 \\ \text{s.t.} & x_1 + 2x_2 \leq 6 \\ & 2x_1 + x_2 \leq 8 \\ & x_i \geq 0 \quad \forall i = 1, 2 \end{array}$$

and its standard form

$$\begin{array}{ll} \max & 2x_1 + 3x_2 \\ \text{s.t.} & x_1 + 2x_2 + x_3 = 6 \\ & 2x_1 + x_2 + x_4 = 8 \\ & x_i \geq 0 \quad \forall i = 1, \dots, 4. \end{array}$$

System of equalities

- ▶ We need to keep track of the **objective value**.
 - ▶ We want to keep improving our solution.
 - ▶ We will use $z = 2x_1 + 3x_2$ to denote the objective value.
 - ▶ The objective value will sometimes be called **the z value**.
- ▶ Once we keep in mind that (1) we are maximizing z and (2) all variables (except z) must be nonnegative, the standard form is nothing but a system of three equalities:

$$\begin{array}{rcccccccl}
 z & - & 2x_1 & - & 3x_2 & & & = & 0 \\
 & & x_1 & + & 2x_2 & + & x_3 & = & 6 \\
 & & 2x_1 & + & x_2 & & & + & x_4 = 8.
 \end{array}$$

- ▶ Note that $z = 2x_1 + 3x_2$ is expressed as $z - 2x_1 - 3x_2 = 0$.
- ▶ This “constraint” (which actually represents the objective function) will be called the 0th constraint.
- ▶ We will repeatedly solve the system.

An initial bfs

- ▶ To start, we need to first have an **initial bfs**.
- ▶ Investigate the system in details:

$$\begin{array}{rclclclcl}
 z & - & 2x_1 & - & 3x_2 & & & = & 0 \\
 & & x_1 & + & 2x_2 & + & x_3 & = & 6 \\
 & & 2x_1 & + & x_2 & & & + & x_4 = 8.
 \end{array}$$

- ▶ Selecting x_3 and x_4 definitely works!
- ▶ In the system, these two columns form an **identity matrix**: $A_B = I$.⁴
- ▶ Moreover, in a standard form LP, the RHS b are nonnegative.
- ▶ Therefore, $x_B = A_B^{-1}b = Ib = b \geq 0$.

⁴For most LPs, such an identity matrix does not exist. We will see how to deal with this situation.

Improving the current bfs

$$\begin{array}{rclclclcl}
 z & - & 2x_1 & - & 3x_2 & & & = & 0 \\
 & & x_1 & + & 2x_2 & + & x_3 & = & 6 \\
 & & 2x_1 & + & x_2 & & & + & x_4 = 8.
 \end{array}$$

- ▶ Let us start from $x^1 = (0, 0, 6, 8)$ and $z_1 = 0$.
- ▶ To move, let's choose a nonbasic variable to enter. x_1 or x_2 ?
 - ▶ The **0th constraints** tells us that entering either variable makes z larger: When one goes up, z goes up to maintain the equality.
 - ▶ For no reason, let's choose x_1 to enter.
- ▶ When to stop?
 - ▶ Now x_1 goes up from 0.
 - ▶ $(0, 0, 6, 8) \rightarrow (1, 0, 5, 6) \rightarrow (2, 0, 4, 4) \rightarrow \dots$. Note that x_2 remains 0.
 - ▶ We will stop at $(4, 0, 2, 0)$, i.e., when x_4 becomes 0.
 - ▶ This is indicated by the **ratio** of the **RHS** and **entering column**:
Because $\frac{8}{2} < \frac{6}{1}$, x_4 becomes 0 sooner than x_3 .
- ▶ We move to $x^2 = (4, 0, 2, 0)$ with $z_2 = 8$.

Keep improving the current bfs

$$\begin{array}{rclclclcl}
 z & - & 2x_1 & - & 3x_2 & & & = & 0 \\
 & & x_1 & + & 2x_2 & + & x_3 & = & 6 \\
 & & 2x_1 & + & x_2 & & & + & x_4 = 8.
 \end{array}$$

- ▶ Let's improve $x^2 = (4, 0, 2, 0)$ by moving to the next bfs.
 - ▶ One of x_2 and x_4 may enter. Let's try to enter x_2 .
- ▶ When x_2 goes up and x_4 remains 0:
 - ▶ The 2nd row says x_2 can at most become 8 (and then x_1 becomes 0).
 - ▶ In the 1st row... how will x_1 and x_3 change?
- ▶ **According to constraint 2**, when x_2 goes up by 1 and x_4 remains 0, x_1 should decrease by $\frac{1}{2}$.
 - ▶ Therefore, according to constraint 1, when x_2 goes up by 1 “and” x_1 goes down by $\frac{1}{2}$, x_3 should go down by $\frac{3}{2}$.
 - ▶ Therefore, x_2 can be at most $\frac{4}{3}$. We reach $(\frac{10}{3}, \frac{4}{3}, 0, 0)$.
- ▶ Collectively, we should increase x_2 by $\min\{8, \frac{4}{3}\}$.
 - ▶ The z value becomes $z_3 = \frac{10}{3} \times 2 + \frac{4}{3} \times 3 = \frac{32}{3}$.
 - ▶ It does not becomes $z_2 + \frac{4}{3} \times 3$ as **the basic variable x_1 also changes**.

Keep improving the current bfs

$$\begin{array}{rclclclclcl}
 z & - & 2x_1 & - & 3x_2 & & & & = & 0 \\
 & & x_1 & + & 2x_2 & + & x_3 & & = & 6 \\
 & & 2x_1 & + & x_2 & & & + & x_4 & = & 8.
 \end{array}$$

- ▶ Note that what we did has two flaws.
- ▶ Regarding constraints:
 - ▶ When we increase the nonbasic variable x_2 , it may affect both basic variables x_1 and x_3 .
 - ▶ Because x_3 does not appear in constraint 2, we know how x_1 responds to the change of x_2 .
 - ▶ We need to consider that to see how x_3 responds to the change of x_2 .
- ▶ Regarding the objective function:
 - ▶ When we increase the nonbasic variable x_2 , it affects basic variables x_1 and x_3 .
 - ▶ Because x_1 is in constraint 0, z is affected by both x_1 and x_2 .
- ▶ How to do these calculations with thousands of variables and constraints?

Keep improving the current bfs

$$\begin{array}{rclclclclclcl}
 z & - & 2x_1 & - & 3x_2 & & & & & = & 0 \\
 & & x_1 & + & 2x_2 & + & x_3 & & & = & 6 \\
 & & 2x_1 & + & x_2 & & & + & x_4 & = & 8.
 \end{array}$$

- ▶ An easier way is to **update the system** before the 2nd move.
 - ▶ To make each of rows 1 to n contains **exactly one** basic variable.
 - ▶ To make row 0 contains **no** basic variable.
- ▶ In other words, for the **basic columns**:
 - ▶ We want an **identity matrix** in rows 1 to m .
 - ▶ We want a **zero vector** in row 0.

Improving the current bfs (the 2nd attempt)

- ▶ Recall that for the system

$$\begin{array}{rccccccc} z & - & 2x_1 & - & 3x_2 & & = & 0 \\ & & x_1 & + & 2x_2 & + & x_3 & = & 6 \\ & & 2x_1 & + & x_2 & & & + & x_4 & = & 8, \end{array}$$

we start from $x^1 = (0, 0, 6, 8)$ with $z_1 = 0$.

- ▶ For the basic columns (the 3rd and 4th ones), indeed we have the identity matrix and zeros.
- ▶ Then we know x_1 enters and x_4 leaves.
 - ▶ The basis becomes (x_1, x_3) .
 - ▶ We need to update the system to

$$\begin{array}{rcccl} z & \boxed{} & + & ?x_2 & \boxed{} & + & ?x_4 & = & 0 \\ & & + & ?x_2 & + & x_3 & + & ?x_4 & = & 6 \\ & x_1 & + & ?x_2 & & & + & ?x_4 & = & 8. \end{array}$$

- ▶ How? **Elementary row operations!**

Updating the system

- ▶ Starting from:

$$z - 2x_1 - 3x_2 = 0 \quad (0)$$

$$x_1 + 2x_2 + x_3 = 6 \quad (1)$$

$$2x_1 + x_2 + x_4 = 8. \quad (2)$$

- ▶ Multiply (2) by $\frac{1}{2}$: $x_1 - \frac{1}{2}x_2 + \frac{1}{2}x_4 = 4$.
- ▶ Multiply (2) by $-\frac{1}{2}$ and then add it into (1): $\frac{3}{2}x_2 + x_3 - \frac{1}{2}x_4 = 2$.
- ▶ Multiply (2) by 1 and then add it into (0): $z - 2x_2 + x_4 = 8$.
- ▶ Collectively, the system becomes

$$z - 2x_2 + x_4 = 8 \quad (0)$$

$$\frac{3}{2}x_2 + x_3 - \frac{1}{2}x_4 = 2 \quad (1)$$

$$x_1 + \frac{1}{2}x_2 + \frac{1}{2}x_4 = 4. \quad (2)$$

- ▶ Updating the system also gives us the objective value $z_2 = 8$ and the current bfs $x^2 = (4, 0, 2, 0)$.

Improving the current bfs (finally!)

- Given the updated system

$$z - 2x_2 + x_4 = 8 \quad (0)$$

$$\frac{3}{2}x_2 + x_3 - \frac{1}{2}x_4 = 2 \quad (1)$$

$$x_1 + \frac{1}{2}x_2 + \frac{1}{2}x_4 = 4, \quad (2)$$

we now know how to do the next iteration.

- We are at $x^2 = (4, 0, 2, 0)$ with $z_2 = 8$.
- One of x_2 and x_4 may enter.
- If x_2 enters, z will go up. Good!
- If x_4 enters, z will go down. Bad.
- Let x_2 enter:
 - Row 1: When x_2 goes up, x_3 goes down. x_2 can be as large as $\frac{2}{3/2} = \frac{4}{3}$.
 - Row 2: When x_2 goes up, x_1 goes down. x_2 can be as large as $\frac{4}{1/2} = 8$.
 - So x_3 becomes 0 sooner than x_1 . x_3 leaves the basis.
- The basic variables become x_1 and x_2 . Let's update again.

Improving once more

- Given the system

$$z \quad - \quad 2x_2 \quad + \quad x_4 = 8 \quad (0)$$

$$\quad \quad \quad \frac{3}{2}x_2 + x_3 - \frac{1}{2}x_4 = 2 \quad (1)$$

$$x_1 + \frac{1}{2}x_2 \quad + \quad \frac{1}{2}x_4 = 4, \quad (2)$$

we now need to update it to fit the new basis (x_1, x_2) .

- Multiply (1) by $\frac{2}{3}$: $x_2 + \frac{2}{3}x_3 - \frac{1}{3}x_4 = \frac{4}{3}$.
 - Multiply (the updated) (1) by $-\frac{1}{2}$ and add it to (2).
 - Multiply (the updated) (1) by 2 and add it to (0).
- We get

$$z \quad + \quad \frac{4}{3}x_3 + \frac{1}{3}x_4 = \frac{32}{3} \quad (0)$$

$$\quad \quad \quad x_2 + \frac{2}{3}x_3 - \frac{1}{3}x_4 = \frac{4}{3} \quad (1)$$

$$x_1 \quad - \quad \frac{1}{3}x_3 + \frac{2}{3}x_4 = \frac{10}{3}. \quad (2)$$

No more improvement!

- ▶ The system

$$z \qquad \qquad \qquad + \frac{4}{3}x_3 \quad + \frac{1}{3}x_4 = \frac{32}{3} \quad (0)$$

$$x_2 \quad + \frac{2}{3}x_3 \quad - \frac{1}{3}x_4 = \frac{4}{3} \quad (1)$$

$$x_1 \quad \quad - \frac{1}{3}x_3 \quad + \frac{2}{3}x_4 = \frac{10}{3} \quad (2)$$

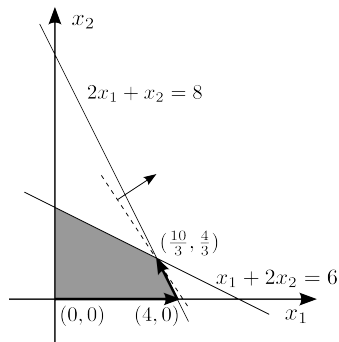
tells us that the new bfs is $x^3 = (\frac{10}{3}, \frac{4}{3}, 0, 0)$ with $z_3 = \frac{32}{3}$.

- ▶ Updating the system also gives us the new bfs and its objective value.
- ▶ Now... no more improvement is needed!
 - ▶ Entering x_3 makes things worse (z must go down).
 - ▶ Entering x_4 also makes things worse.
- ▶ x^3 is an optimal solution.⁵ We are done!

⁵This is indeed true, though a rigorous proof is omitted.

Visualizing the iterations

- ▶ Let's visualize this example and relate bfs with extreme points.
 - ▶ The initial bfs corresponds to $(0, 0)$.
 - ▶ After one iteration, we move to $(4, 0)$.
 - ▶ After two iterations, we move to $(\frac{10}{3}, \frac{4}{3})$, which is optimal.
- ▶ Please note that we move along edges to search among extreme points!



Summary

- ▶ To run the simplex method:
 - ▶ Find an initial bfs with its basis.⁶
 - ▶ Among those nonbasic variables with positive coefficients in the 0th row,⁷ choose one to enter.⁸
 - ▶ If there is none, terminate and report the current bfs as optimal.
 - ▶ According to the ratios from the entering and RHS columns, decide which basic variable should leave.⁹
 - ▶ Find a new basis.
 - ▶ Make the system fit the requirements for basic columns:
 - ▶ Identity matrix in constraints (1st to m th row).
 - ▶ Zeros in the objective function (0th row).
 - ▶ Repeat.

⁶How to find one?

⁷Positive coefficients for a minimization problem; negative for maximization.

⁸What if there are multiple?

⁹What if there is a tie? What if the denominator is 0 or negative?

Road map

- ▶ The standard form.
- ▶ Basic solutions.
- ▶ The simplex method.
- ▶ **The tableau representation.**
- ▶ Unbounded LPs.
- ▶ Infeasible LPs.

The tableau representation

- ▶ We typically omit variables when updating those systems.
- ▶ We organize coefficients into **tableaus**.
 - ▶ As the column with z never changes, we do not include it in a tableau.
- ▶ For our example, the initial system

$$\begin{array}{rclclclclcl} z & - & 2x_1 & - & 3x_2 & & & = & 0 \\ & & x_1 & + & 2x_2 & + & x_3 & = & 6 \\ & & 2x_1 & + & x_2 & & & + & x_4 = 8. \end{array}$$

can be expressed as

$$\begin{array}{cccc|c} -2 & -3 & 0 & 0 & 0 \\ \hline 1 & 2 & 1 & 0 & x_3 = 6 \\ 2 & 1 & 0 & 1 & x_4 = 8 \end{array}$$

- ▶ The basic columns have zeros in the 0th row and an identity matrix in the other rows.
- ▶ The identity matrix associates each row with a basic variable.
- ▶ A negative number in the 0th row of a nonbasic column means that variable can enter.

Using tableaus rather than systems

$$\begin{array}{rclclcl} z & - & 2x_1 & - & 3x_2 & & = & 0 \\ & & x_1 & + & 2x_2 & + & x_3 & = & 6 \\ & & 2x_1 & + & x_2 & & & + & x_4 & = & 8 \end{array}$$

$$\begin{array}{cccc|c} -2 & -3 & 0 & 0 & 0 \\ \hline 1 & 2 & 1 & 0 & x_3 = 6 \\ \boxed{2} & 1 & 0 & 1 & x_4 = 8 \end{array}$$

↓

$$\begin{array}{rclclcl} z & & - & 2x_2 & & + & x_4 & = & 8 \\ & & + & \frac{3}{2}x_2 & + & x_3 & - & \frac{1}{2}x_4 & = & 2 \\ x_1 & + & \frac{1}{2}x_2 & & & + & \frac{1}{2}x_4 & = & 4 \end{array}$$

$$\begin{array}{cccc|c} 0 & -2 & 0 & 1 & 8 \\ \hline 0 & \boxed{\frac{3}{2}} & 1 & -\frac{1}{2} & x_3 = 2 \\ 1 & \frac{1}{2} & 0 & \frac{1}{2} & x_1 = 4 \end{array}$$

↓

$$\begin{array}{rclclcl} z & & & + & \frac{4}{3}x_3 & + & \frac{1}{3}x_4 & = & \frac{32}{3} \\ & & x_2 & + & \frac{2}{3}x_3 & - & \frac{1}{3}x_4 & = & \frac{4}{3} \\ x_1 & & & - & \frac{1}{3}x_3 & + & \frac{2}{3}x_4 & = & \frac{10}{3} \end{array}$$

$$\begin{array}{cccc|c} 0 & 0 & \frac{4}{3} & \frac{1}{3} & \frac{32}{3} \\ \hline 0 & 1 & \frac{2}{3} & -\frac{1}{3} & x_2 = \frac{4}{3} \\ 1 & 0 & -\frac{1}{3} & \frac{2}{3} & x_1 = \frac{10}{3} \end{array}$$

The second example

- Consider another example:

$$\begin{array}{ll}
 \max & x_1 \\
 \text{s.t.} & 2x_1 - x_2 \leq 4 \\
 & 2x_1 + x_2 \leq 8 \\
 & x_2 \leq 3 \\
 & x_i \geq 0 \quad \forall i = 1, 2.
 \end{array}$$

- The standard form is

$$\begin{array}{ll}
 \max & x_1 \\
 \text{s.t.} & 2x_1 - x_2 + x_3 = 4 \\
 & 2x_1 + x_2 + x_4 = 8 \\
 & x_2 + x_5 = 3 \\
 & x_i \geq 0 \quad \forall i = 1, \dots, 5.
 \end{array}$$

The first iteration

- We prepare the initial tableau. We have $x^1 = (0, 0, 4, 8, 3)$ and $z_1 = 0$.

$$\begin{array}{ccccc|c}
 -1 & 0 & 0 & 0 & 0 & 0 \\
 \hline
 2 & -1 & 1 & 0 & 0 & x_3 = 4 \\
 2 & 1 & 0 & 1 & 0 & x_4 = 8 \\
 0 & 1 & 0 & 0 & 1 & x_5 = 3
 \end{array}$$

- For this **maximization** problem, we look for **negative** numbers in the 0th row. Therefore, x_1 enters.
- Those numbers in the 0th row are called **reduced costs**.
 - The 0th row is $z - x_1 = 0$. Increasing x_1 can increase z .
- “Dividing the RHS column by the entering column” tells us that x_3 should leave (it has the minimum ratio).¹⁰
- This is called the **ratio test**. We **always** look for the smallest ratio.

¹⁰The 0 in the 3rd row means that increasing x_1 does not affect x_5 .

The first iteration

- x_1 enters and x_3 leaves. The next tableau is found by **pivoting** at 2:

$$\begin{array}{ccccc|c} -1 & 0 & 0 & 0 & 0 & 0 \\ \hline \boxed{2} & -1 & 1 & 0 & 0 & x_3 = 4 \\ 2 & 1 & 0 & 1 & 0 & x_4 = 8 \\ 0 & 1 & 0 & 0 & 1 & x_5 = 3 \end{array} \rightarrow \begin{array}{ccccc|c} 0 & -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 2 \\ \hline 1 & -\frac{1}{2} & \frac{1}{2} & 0 & 0 & x_1 = 2 \\ 0 & 2 & -1 & 1 & 0 & x_4 = 4 \\ 0 & 1 & 0 & 0 & 1 & x_5 = 3 \end{array}$$

- The new bfs is $x^2 = (2, 0, 0, 4, 3)$ with $z_2 = 2$.
- Continue?
- There is a negative reduced cost in the 2nd column: x_2 enters.
- Ratio test:
- That $-\frac{1}{2}$ in the 1st row shows that increasing x_2 makes x_1 larger. Row 1 does not participate in the ratio test.
- For rows 2 and 3, row 2 wins (with a smaller ratio).

The second iteration

- ▶ x_2 enters and x_4 leaves. We pivot at 2.
- ▶ The second iteration is

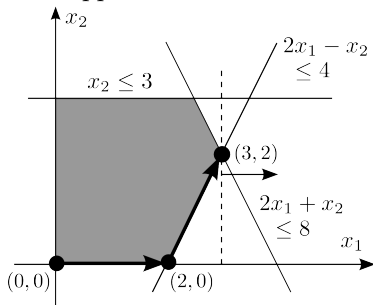
$$\begin{array}{ccccc|c}
 0 & -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 2 \\
 \hline
 1 & -\frac{1}{2} & \frac{1}{2} & 0 & 0 & x_1 = 2 \\
 0 & \boxed{2} & -1 & 1 & 0 & x_4 = 4 \\
 0 & 1 & 0 & 0 & 1 & x_5 = 3
 \end{array}
 \rightarrow
 \begin{array}{ccccc|c}
 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 3 \\
 \hline
 1 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & x_1 = 3 \\
 0 & 1 & -\frac{1}{2} & \frac{1}{2} & 0 & x_2 = 2 \\
 0 & 0 & \frac{1}{2} & -\frac{1}{2} & 1 & x_5 = 1
 \end{array}$$

- ▶ The third bfs is $x^3 = (3, 2, 0, 0, 1)$ with $z_3 = 3$.
 - ▶ It is optimal (why?).
 - ▶ Typically we write the optimal solution we find as x^* and optimal objective value as z^* .

Visualizing the solution process

- ▶ The three basic feasible solutions we obtain are
 - ▶ $x^1 = (0, 0, 4, 8, 3)$.
 - ▶ $x^2 = (2, 0, 0, 4, 3)$.
 - ▶ $x^3 = x^* = (3, 2, 0, 0, 1)$.

Do they fit our graphical approach?



Road map

- ▶ The standard form.
- ▶ Basic solutions.
- ▶ The simplex method.
- ▶ The tableau representation.
- ▶ **Unbounded LPs.**
- ▶ Infeasible LPs.

Identifying unboundedness

- ▶ When is an LP **unbounded**?
- ▶ An LP is unbounded if:
 - ▶ There is an improving direction.
 - ▶ Along that direction, we may move forever.
- ▶ When we run the simplex method, this can be easily checked in a simplex tableau.
- ▶ Consider the following example:

$$\begin{array}{ll}\max & x_1 \\ \text{s.t.} & x_1 - x_2 \leq 1 \\ & 2x_1 - x_2 \leq 4 \\ & x_i \geq 0 \quad \forall i = 1, 2.\end{array}$$

Unbounded LPs

- The standard form is:

$$\begin{array}{llllllll}
 \max & x_1 & & & & & & \\
 \text{s.t.} & x_1 & - & x_2 & + & x_3 & & = & 1 \\
 & 2x_1 & - & x_2 & & & + & x_4 & = & 4 \\
 & x_i & \geq & 0 & \forall i = 1, \dots, 4.
 \end{array}$$

- The first iteration:

$$\begin{array}{cccc|c}
 -1 & 0 & 0 & 0 & 0 \\
 \hline
 \boxed{1} & -1 & 1 & 0 & x_3 = 1 \\
 2 & -1 & 0 & 1 & x_4 = 4
 \end{array}
 \rightarrow
 \begin{array}{cccc|c}
 0 & -1 & 1 & 0 & 1 \\
 \hline
 1 & -1 & 1 & 0 & x_1 = 1 \\
 0 & 1 & -2 & 1 & x_4 = 2
 \end{array}$$

Unbounded LPs

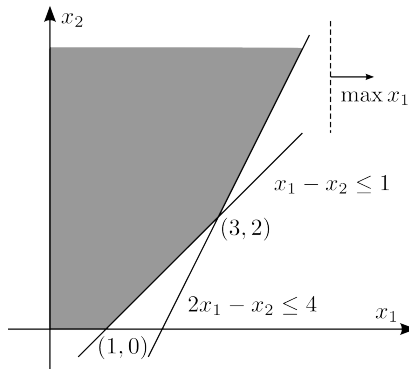
- The second iteration:

$$\begin{array}{cccc|c}
 0 & -1 & 1 & 0 & 1 \\
 \hline
 1 & -1 & 1 & 0 & x_1 = 1 \\
 0 & \boxed{1} & -2 & 1 & x_4 = 2
 \end{array}
 \rightarrow
 \begin{array}{cccc|c}
 0 & 0 & -1 & 1 & 3 \\
 \hline
 1 & 0 & -1 & 1 & x_1 = 3 \\
 0 & 1 & -2 & 1 & x_2 = 2
 \end{array}$$

- How may we do the third iteration? The **ratio test** fails!
- Only rows with positive denominators participate in the ratio test.
 - Now all the denominators are nonpositive! Which variable to leave?
- No one should leave: Increasing x_3 makes x_1 and x_2 become larger.
- Row 1: $x_1 - x_3 + x_4 = 3$.
 - Row 2: $x_2 - 2x_3 + x_4 = 2$.
- The direction is thus an **unbounded improving direction**.

Unbounded improving directions

- At $(3, 2)$, when we enter x_3 , we move along the rightmost edge. Geometrically, both nonbinding constraints $x_1 \geq 0$ and $x_2 \geq 0$ are “behind us”.



Detecting unbounded LPs

- ▶ For a **minimization** LP, whenever we see **any** column in **any** tableau

$$\begin{array}{c|c}
 \bar{c}_j & \\
 \hline
 d_1 & \\
 \vdots & \\
 d_m &
 \end{array}$$

such that $\bar{c}_j > 0$ and $d_i \leq 0$ for all $i = 1, \dots, m$, we may stop and conclude that this LP is unbounded.

- ▶ $\bar{c}_j > 0$: This is an improving direction.
- ▶ $d_i \leq 0$ for all $i = 1, \dots, m$: This is an unbounded direction.
- ▶ What is the unbounded condition for a **maximization** problem?

Road map

- ▶ The standard form.
- ▶ Basic solutions.
- ▶ The simplex method.
- ▶ The tableau representation.
- ▶ Unbounded LPs.
- ▶ **Infeasible LPs.**

Feasibility of an LP

- ▶ When an LP

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0\end{array}$$

satisfies $b \geq 0$, finding a bfs for its standard form

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax + Iy = b \\ & x, y \geq 0,\end{array}$$

is trivial.

- ▶ We may form a feasible basis with all the slack variables y .
- ▶ What if there are some “=” or “ $\geq$ ” constraints?

Feasibility of an LP

► For example, given an LP

$$\begin{array}{ll}
 \min & x_1 \\
 \text{s.t.} & x_1 + x_2 - x_3 + x_4 \geq 10 \\
 & 3x_1 + 2x_2 + 9x_3 - x_4 = 10 \\
 & x_1 - 8x_2 + 2x_3 - 6x_4 \leq 10 \\
 & x_i \geq 0 \quad \forall i = 1, \dots, 4
 \end{array}$$

whose standard form is

$$\begin{array}{ll}
 \min & x_1 \\
 \text{s.t.} & x_1 + x_2 - x_3 + x_4 - x_5 = 10 \\
 & 3x_1 + 2x_2 + 9x_3 - x_4 = 10 \\
 & x_1 - 8x_2 + 2x_3 - 6x_4 + x_6 = 10 \\
 & x_i \geq 0 \quad \forall i = 1, \dots, 6,
 \end{array}$$

it is nontrivial to find a feasible basis (if there is one).

The two-phase implementation

- ▶ To find an initial bfs (or show that there is none), we may apply the **two-phase implementation**.
- ▶ Given a standard form LP (P) , we construct a **phase-I LP** (Q) :¹¹

$$(P) \quad \begin{array}{ll} \min & c^T x \\ \text{s.t.} & Ax = b \\ & x \geq 0 \end{array}$$

$$(Q) \quad \begin{array}{ll} \min & 1^T y \\ \text{s.t.} & Ax + Iy = b \\ & x, y \geq 0. \end{array}$$

- ▶ (Q) has a trivial bfs $(x, y) = (0, b)$, so we can apply the simplex method on (Q) . But so what?

Proposition 2

(P) is feasible if and only if (Q) has an optimal bfs $(x, y) = (\bar{x}, 0)$. In this case, $\bar{x}$ is a bfs of (P) .

¹¹Even if in (P) we have a maximization objective function, (Q) is still the same.

The two-phase implementation

- ▶ After we solve (Q) , either we know (P) is infeasible or we have a feasible basis of (P) .
- ▶ In the latter case, we can recover the objective function of the original (P) to get a **phase-II LP**.
 - ▶ “The phase-II LP” is nothing but the original (P) .
 - ▶ Phase I for a **feasible** solution and phase II for an **optimal** solution.
- ▶ Regarding those added variables:
 - ▶ They are **artificial variables** and have no physical meaning. They are created only for checking feasibility.
 - ▶ If a constraint already has a variable that can be included in a trivial basis, we do not need to add an artificial variable in that constraint.
 - ▶ This happens to those “ $\leq$ ” constraints (if the RHS is nonnegative).
- ▶ We then **adjust** the tableau according to the initial basis and **continue** applying the simplex method on the phase-II LP.

Example 1: Phase I

- Consider an LP

$$\begin{array}{llll} \max & x_1 & + & x_2 \\ \text{s.t.} & 2x_1 & + & x_2 \geq 6 \\ & x_1 & + & 2x_2 \leq 6 \\ & x_i & \geq 0 & \forall i = 1, 2. \end{array}$$

which has no trivial bfs (due to the “ $\geq$ ” constraint).

- Its Phase-I standard form LP is

$$\begin{array}{llllllll} \min & & & & & & & x_5 \\ \text{s.t.} & 2x_1 & + & x_2 & - & x_3 & & + & x_5 = 6 \\ & x_1 & + & 2x_2 & & & + & x_4 & = 6 \\ & x_i & \geq 0 & \forall i = 1, \dots, 5. \end{array}$$

- We need only one artificial variable x_5 . x_3 and x_4 are slack variables.

Example 1: preparing the initial tableau

- Let's try to solve the Phase-I LP. First, let's prepare the initial tableau:

$$\begin{array}{ccccc|c}
 0 & 0 & 0 & 0 & -1 & 0 \\
 \hline
 2 & 1 & -1 & 0 & 1 & x_5 = 6 \\
 1 & 2 & 0 & 1 & 0 & x_4 = 6
 \end{array}$$

- Is this a valid tableau? No!
- For all basic columns (in this case, columns 4 and 5), the 0th row should contain 0.
 - So we need to first **adjust the 0th row** with elementary row operations.

Example 1: preparing the initial tableau

- Let's adjust row 0 by adding row 1 to row 0.

$$\begin{array}{ccccc|c}
 0 & 0 & 0 & 0 & -1 & 0 \\
 \hline
 2 & 1 & -1 & 0 & 1 & x_5 = 6 \\
 1 & 2 & 0 & 1 & 0 & x_4 = 6
 \end{array}
 \xrightarrow{\text{adjust}}
 \begin{array}{ccccc|c}
 2 & 1 & -1 & 0 & 0 & 6 \\
 \hline
 2 & 1 & -1 & 0 & 1 & x_5 = 6 \\
 1 & 2 & 0 & 1 & 0 & x_4 = 6
 \end{array}$$

- Now we have a valid initial tableau to start from!
- The current bfs is $x^0 = (0, 0, 0, 6, 6)$, which corresponds to an **infeasible** solution to the original LP.
- We know this because there are positive artificial variables.

Example 1: solving the Phase-I LP

- ▶ Solving the Phase-I LP takes only one iteration:

$$\begin{array}{c|c}
 2 & 1 & -1 & 0 & 0 & 6 \\
 \hline
 \boxed{2} & 1 & -1 & 0 & 1 & x_5 = 6 \\
 1 & 2 & 0 & 1 & 0 & x_4 = 6
 \end{array}
 \rightarrow
 \begin{array}{c|c}
 0 & 0 & 0 & 0 & 0 \\
 \hline
 1 & \frac{1}{2} & -\frac{1}{2} & 0 & x_1 = 3 \\
 0 & \frac{3}{2} & \frac{1}{2} & 1 & x_4 = 3
 \end{array}$$

- ▶ Whenever an artificial variable leaves the basis, we will not need to enter it again. Therefore, we may remove that column to save calculations.
- ▶ As we can remove all artificial variables, the original LP is feasible.
- ▶ A feasible basis for the original LP is (x_1, x_4) .

Example 1: solving the Phase-II LP

- ▶ Now let's construct the Phase-II LP.
- ▶ Step 1: put the original objective function “ $\max x_1 + x_2$ ” back:

$$\begin{array}{cccc|c}
 -1 & -1 & 0 & 0 & 0 \\
 \hline
 1 & \frac{1}{2} & -\frac{1}{2} & 0 & x_1 = 3 \\
 0 & \frac{3}{2} & \frac{1}{2} & 1 & x_4 = 3
 \end{array}$$

- ▶ Is this a valid tableau? No!
 - ▶ Column 1, which should be basic, contains a nonzero number in the 0th row. It must be adjusted to 0.
- ▶ Before we run iterations, let's adjust the 0th row again.

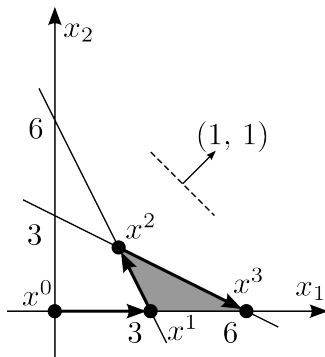
Example 1: solving the Phase-II LP

- Let's fix the 0th row and then run two iterations.

$$\begin{array}{c}
 \begin{array}{cccc|c}
 -1 & -1 & 0 & 0 & 0 \\
 \hline
 1 & \frac{1}{2} & -\frac{1}{2} & 0 & x_1 = 3 \\
 0 & \frac{3}{2} & \frac{1}{2} & 1 & x_4 = 3
 \end{array}
 \quad \xrightarrow{\text{adjust}} \quad
 \begin{array}{cccc|c}
 0 & -\frac{1}{2} & -\frac{1}{2} & 0 & 3 \\
 \hline
 1 & \frac{1}{2} & -\frac{1}{2} & 0 & x_1 = 3 \\
 0 & \boxed{\frac{3}{2}} & \frac{1}{2} & 1 & x_4 = 3
 \end{array} \\
 \\
 \begin{array}{c}
 \rightarrow \quad
 \begin{array}{cccc|c}
 0 & 0 & -\frac{1}{3} & \frac{1}{3} & 4 \\
 \hline
 1 & 0 & -\frac{2}{3} & -\frac{1}{3} & x_1 = 2 \\
 0 & 1 & \boxed{\frac{1}{3}} & \frac{2}{3} & x_2 = 2
 \end{array}
 \quad \rightarrow \quad
 \begin{array}{cccc|c}
 0 & 1 & 0 & 1 & 6 \\
 \hline
 1 & 2 & 0 & 1 & x_1 = 6 \\
 0 & 3 & 1 & 2 & x_3 = 6
 \end{array}
 \end{array}
 \end{array}$$

- The optimal bfs is $(6, 0, 6, 0)$.

Example 1: visualization



- ▶ x^0 is infeasible (the artificial variable x_5 is positive).
- ▶ x^1 is the initial bfs (as a result of Phase I).
- ▶ x^3 is the optimal bfs (as a result of Phase II).